

Society Maintenance Tracker

Objective:

- Apartment societies handle a steady stream of maintenance complaints, but without a proper system the admin has no way to see what is pending, what is overdue, or which issues keep coming back. Residents, meanwhile, have no visibility into progress. Build a platform where residents raise and track complaints with supporting photos, the admin manages them through a clear workflow with priorities, and everyone stays informed through a notice board and email updates.

Scope of Work:

- **Input:** Resident complaints with photos, admin status and priority updates, admin notices
- **Output:** Tracked complaints with history, a notice board, and email updates to residents
- Resident can register, log in, and raise a complaint with a category, description, and optional photo
- Resident can view all their complaints with the full status history of each
- Admin can view all complaints, filter by category, status, or date, and set a priority (Low, Medium, High)
- Admin updates complaint status (Open, In Progress, Resolved); each change is recorded with a timestamp and an optional note
- Once a complaint is marked Resolved, it is closed
- Admin can flag a complaint as overdue if it stays open beyond a configurable number of days, and these surface at the top of the admin view
- Admin can post notices to a notice board, optionally marking a notice as important so it is pinned to the top
- Resident receives an email when their complaint status changes and when a new important notice is posted
- Admin can view a simple dashboard: total complaints by status, by category, and count of overdue complaints

Technical Expectations:

- Backend API, Frontend, and Database with role-based auth (resident, admin)
- Complaint status lifecycle with a recorded history of every change (timestamp, actor, note)
- Photo upload handling for complaints
- Overdue detection based on a configurable threshold
- Notice board with pinned important notices
- Email integration for status changes and important notices (any free tier service)

Deliverables:

1. Zip file with complete source code
2. README with setup guide, .env.example, API docs, and database schema
3. Hosted application URL (Vercel, Render, Railway, or similar)
4. System design write-up (800 words max) covering your complaint history model, overdue detection, photo handling, and notification flow

Evaluation Focus:

- Complaint lifecycle and status history design

- Overdue detection and priority handling
- Photo upload and notice board design
- Dashboard and reporting
- Database schema, API design, and documentation